

Full-Stack Developer Intern (Assignment)

Creator Intelligence Module (PRAJA)

Problem Statement

Build a working full-stack slice that takes a social media creator handle (Instagram), collects their public data, computes quantitative metrics, classifies the comments on their most recent 12 collaboration and personal posts, estimates aggregate audience characteristics, stores it in a clean schema, and presents it on a dashboard. Use APIs where available (Public Crawl/Instagram Graph API / Business Discovery etc.)

What to Build

Creator-level data & metrics:

- Handle, display name, profile picture, bio, external link, verified status, follower count, following count, post/video count.
- Per recent post (last ~12): likes, comments, shares, saves where available. Computed: average likes, average comments, average shares, average saves, most used hashtags and engagement rate (define and justify your formula).
- Identify the most recent collaboration/sponsored post and compute its engagement metrics.

Comment classification (on their 12 recent post both collab and personal ones — classify the comment text):

Bucket	Values
Authenticity	Genuine vs Spam (emoji-only, repeated text, link-drops, follow-for-follow, bot handles) — report genuine ratio
Bot-likelihood	Human / Likely-bot / Uncertain
Political Inclination	Positive / Neutral / Negative (which Party has more inclination)
Relevance	On-topic (post/brand/product) vs Off-topic / generic
Type	Praise / Question / Criticism / Tag-a-friend / Sales-or-promo / Other
Language / script	English / Hindi / Hinglish / regional — flag ambiguous Hinglish for manual review

Aggregate follower estimation wrt to DP and Bio (audience-wide % distributions, each with a confidence level — High / Medium / Low):

Aggregate	Buckets / values
Gender split	Female % / Male % / Undisclosed-Unknown % (report all three)
Interest cohorts	Audience share across the niche cohorts below (if applicable)
Political Inclination	Positive / Neutral / Negative (which Party has more inclination)

Concentrated Niche / Interest Cohorts

Interest cohort	Includes
Beauty & personal care	makeup, skincare, grooming
Fashion & lifestyle	apparel, styling, daily-life vlogging
Fitness & wellness	gym, yoga, nutrition, sport
Food & cooking	recipes, food reviews, regional cuisine
Tech & gadgets	reviews, how-to, unboxing
Travel	destinations, regional tourism
Entertainment & comedy	skits, memes, reactions
Education & knowledge	exam prep, skilling, explainers
Parenting & family	homemaking, kids, family vlogs
Devotional / spiritual	faith-based lifestyle content

Deadline: 21/06/2026